Cross-Site Transfer Learning versus Local Models for Short-Term Photovoltaic Forecasting under Data Scarcity: A Leave-One-Plant-Out Benchmark on 94 Chilean Plants

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***Abstract*—Newly commissioned photovoltaic (PV) plants lack the local generation history that conventional forecasting models require, a *Cold-Start* problem that translates into operational and financial risk for grid operators and small distributed generators. We investigate whether a single *Cross-Site* deep-learning model, pretrained on many consolidated plants, can forecast a brand-new plant it has never observed, and how it compares against strictly local baselines. We benchmark eight configurations—global Temporal Fusion Transformer (TFT), Informer, N-HiTS, LSTM and a global XGBoost against strictly local XGBoost, LSTM and N-HiTS—under a Leave-One-Plant-Out (LOPO) protocol that hides the target plant during training and seeds inference with a purely synthetic context derived from a regional efficiency prior computed only on the training plants. Evaluation spans 24-hour and autoregressive 7-day horizons over 94 Chilean plants (2014–2024), with point (RMSE, sMAPE, rRMSE) and probabilistic (90 % interval coverage, pinball loss) metrics, stratified by season. The results deliver a nuanced verdict. In absolute accuracy the hypothesis is rejected: a mature local XGBoost (62 % median rRMSE, weekly roll-out) beats the best global model (global XGBoost, 95 %). Yet at matched deep architecture, global transfer is competitive—global LSTM (118 %) outperforms local LSTM (147 %) despite never seeing the target. Winter is systematically the hardest season, and the deep networks' probabilistic intervals are markedly miscalibrated under zero-shot transfer. The Cross-Site approach is thus confirmed as operationally viable precisely where the local paradigm cannot exist—the first hours after grid connection.**

***Index Terms*—photovoltaic forecasting, transfer learning, cold-start, deep learning, Temporal Fusion Transformer, probabilistic forecasting, leave-one-plant-out.**

I. Introduction

The accelerated deployment of solar photovoltaic (PV) generation confronts grid operators with a recurrent operational barrier: a plant that has just been connected to the network has no operating history from which to learn its production dynamics. Classical PV forecasting pipelines train a dedicated model per plant on months or years of local data [1]; applied to a freshly commissioned facility they degrade sharply or cannot be trained at all. This is the *Cold-Start* problem, borrowed from recommender systems [2]: reliable inference is required exactly when site-specific evidence is absent. The resulting uncertainty inflates balancing costs and financial exposure for system operators and, acutely, for the small distributed generators that dominate the recent capacity additions.

A promising escape is to abandon the one-model-per-plant assumption. Instead of learning in isolation, a single *global* model can be trained across many plants simultaneously, absorbing shared atmospheric-to-power patterns and transferring them to unseen sites [3], [4]. Modern attention-based sequence models—the Temporal Fusion Transformer (TFT) [5], Informer [6]—and efficient hierarchical alternatives such as N-HiTS [7] make this *Cross-Site* strategy technically feasible on multivariate, exogenous-rich PV data. The open question, and the one this work addresses, is empirical and adversarial: *under a strictly zero-shot protocol, does a global Cross-Site model actually beat mature local baselines, or does the transfer premium evaporate once the comparison is made fair?*

Much of the literature reports favourable global-transfer results but rarely combines three demanding conditions at once: (i) a target-plant-blind evaluation with no local fine-tuning, (ii) a direct contrast against mature local baselines under an identical protocol, and (iii) a probabilistic characterization of the forecasts. We close this gap. Our contributions are:

- A reproducible LOPO benchmark of **eight configurations** (five global, three strictly local) over **94 Chilean PV plants**, with formally verified zero-shot integrity (no feature is a function of the target's own generation).
- A **synthetic Cold-Start seeding** mechanism that initializes the inference context from a regional efficiency prior computed exclusively on training plants, enabling inference with zero target history without leakage.
- A **point-and-probabilistic** evaluation stratified by season and by evaluation window, exposing (a) the rejection of the global-superiority hypothesis in absolute terms, (b) its nuance at matched architecture, (c) winter as the hardest regime, and (d) a systematic miscalibration of the deep networks' intervals under transfer.

II. Related Work

PV forecasting. Deep networks consistently surpass classical statistical methods for short-term PV power [8], [9], and on real solar datasets attention-based models improve over classical machine learning at the cost of higher data and compute demand [10]. The TFT has been applied to day-ahead PV forecasting with quantile outputs and interpretable covariate fusion [11]. A recurrent limitation across these works is the assumption of sufficient local history for training.

Global vs. local forecasting. Montero-Manso and Hyndman [3] formalize why a single global model fit across related series can generalize at least as well as many local models, given enough cross-series signal. For intermittent series, however, the balance between local and global models is not settled [12], and simple baselines often rival transformers on standard benchmarks [13]—a caution we find echoed in our PV results.

Transfer learning for PV. Bak et al. [14] pretrain on large multi-plant corpora and fine-tune on the first records of a new site, attenuating error during the first weeks of operation—but still requiring a local fine-tuning phase. Foundation-model approaches push toward pure zero-shot irradiance transfer [15]. Our study differs by enforcing a strict zero-shot regime *and* contrasting it, on the same footing, against mature local baselines—a comparison typically absent from the transfer literature.

Probabilistic forecasting. Quantile and distributional outputs are standard for decision-relevant energy forecasting [16], [17]. We adopt 90 % prediction intervals and evaluate coverage and pinball loss, but crucially examine whether calibration itself survives the zero-shot transfer.

III. Methodology

A. Dataset and Medallion ETL

The corpus comprises hourly generation from 94 PV plants of the Chilean National Electric System (2014–2024), dynamic exogenous meteorology (global horizontal irradiance, temperature, humidity) from the CR2 network, and static geographic attributes (macro-zone, installed capacity). The plants span the country's steep solar gradient and are grouped into four macro-zones from the Atacama desert in the north to the south; this grouping is the static key over which the regional prior is aggregated. A Medallion pipeline (landing → bronze → silver → gold, Fig. 1) parses the raw meteorological and generation CSVs, standardizes them to a canonical $(unique\_id,\, ds,\, y)$ schema, joins them into a single `silver_unified` table (≈4.37 M hourly rows), engineers cyclic calendar features, and imputes *only* exogenous variables (each flagged with an `is_imputed` dummy). Generation y is never imputed: missing hours remain missing, and metrics score real observations only. A numeric, label-encoded `silver_dl` view feeds the deep networks, while the gold layer materializes one held-out test partition per plant for the LOPO evaluation. A single orchestrator runs the pipeline behind a pre-run test gate with idempotent completion markers.

B. LOPO Protocol and Zero-Shot Integrity

Evaluation follows a Leave-One-Plant-Out design. For each target plant, the global models are trained on the full 2014–2024 history of the remaining $N-1$ plants; the target is *completely invisible* during training. Zero-shot integrity is a hard constraint: no predictor may be a function of the target's own y. This is enforced by automated tests and was decisive—an early efficiency prior computed per-row from the target ($\mathrm{corr} = 1.0$ with y) produced a fake ∼0.8 % rRMSE that collapsed to the honest ∼56 % once removed.

C. Synthetic Cold-Start Context

Deep models require a context window to initialize inference, which a new plant cannot provide. We seed it synthetically:

$$\hat{y}_{\text{seed}}(t) = \text{capacity} \cdot PR_{\text{regional}} \cdot s(t), \tag{1}$$

where $s(t)$ is a deterministic solar profile and PR_{regional} is a performance ratio aggregated by macro-zone and season *over the training plants only*. Only known or forecastable exogenous weather and static geography use the target's real values; the generation context is entirely synthetic (Fig. 2).

D. Models and Probabilistic Output

We evaluate five global configurations—TFT [5], Informer [6], N-HiTS [7], LSTM [18] and a global gradient-boosted XGBoost [19]—and three strictly local baselines (XGBoost, LSTM, N-HiTS) trained only on the target's own history with all evaluation windows excluded. The four deep architectures span complementary inductive biases: the TFT combines variable-selection networks with interpretable multi-head attention and gating; the Informer replaces full attention with a ProbSparse mechanism for long sequences; N-HiTS performs hierarchical multi-rate interpolation; and the LSTM is a strong recurrent baseline. Deep models are trained with NeuralForecast in `fp32` on a 6 GB GPU under an epoch-based step budget capped per strategy, with early stopping on a 168-hour validation split; hyperparameters are tuned once per model and cached across LOPO folds (a negligible, documented parameter-level leakage). Every model emits P5/P50/P95 quantiles (90 % intervals), and predictions are physically constrained ($\hat{y} \geq 0$; forced to zero at night).

E. Evaluation Windows and Metrics

Two horizons are scored: *day1* (24 h direct) and *rollout7d* (7 autoregressive days feeding back the predicted median). Because early series often contain a *commissioning ramp* of null generation, we report a window sensitivity analysis (Fig. 2): *Raw* (from the first recorded hour, ramp included), *Operational* (from first sustained production) and four *Seasonal* windows starting in each season, drawn from the plant's own later history. Point metrics are RMSE, sMAPE and rRMSE (normalized by the window's mean hourly generation, which includes null night hours and therefore inflates the index—its value is the *relative* comparison across architectures on identical windows). Probabilistic quality uses 90 % coverage

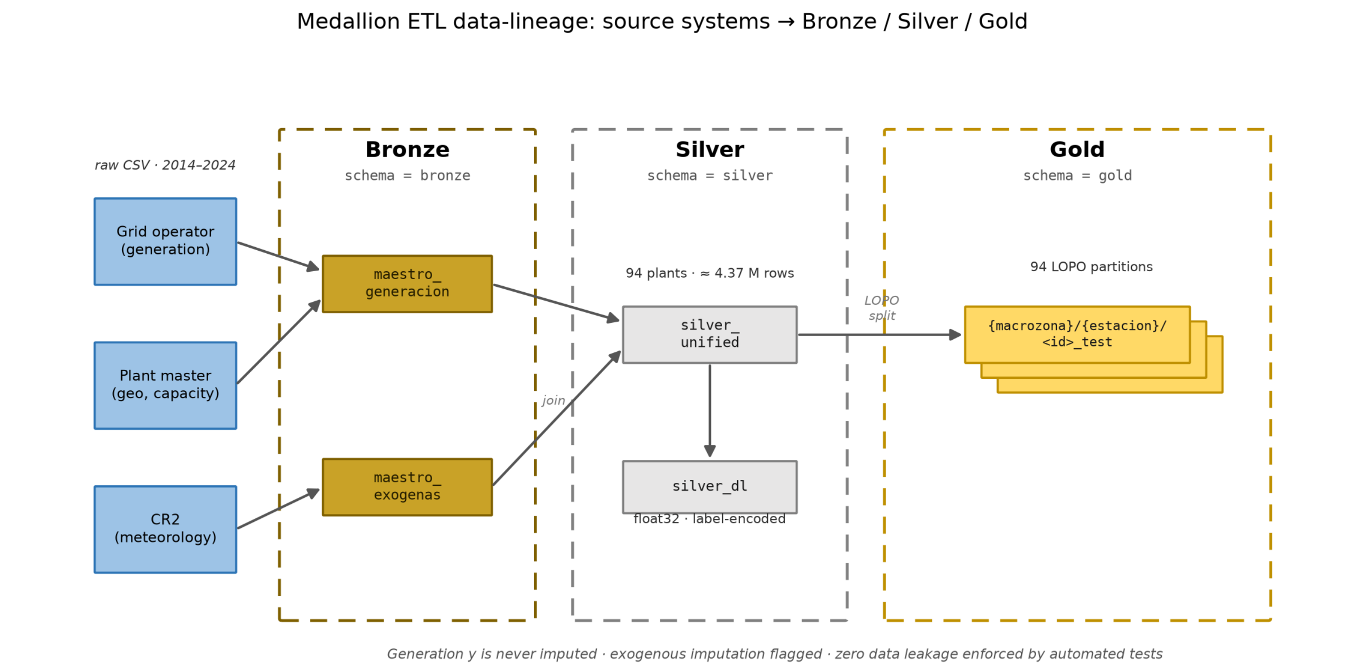

Fig. 1. Medallion ETL data-lineage. Source systems feed two Bronze master tables, joined into `silver_unified` and its numeric `silver_dl` view; the gold layer holds one LOPO test partition per plant. Generation is never imputed, and zero leakage is enforced by automated tests.

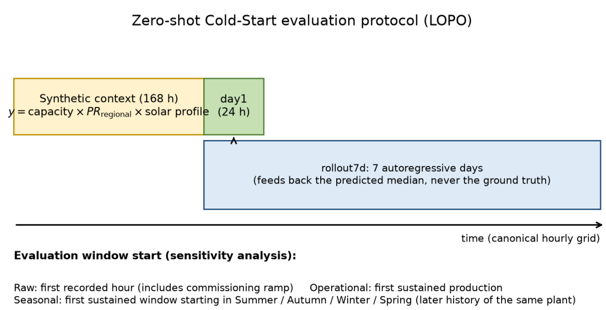

Fig. 2. Zero-shot Cold-Start evaluation protocol. A 168-hour synthetic context seeds the inference; the model then predicts the *day1* (24 h) and *rollout7d* horizons. The window start defines the Raw/Operational/Seasonal sensitivity analysis.

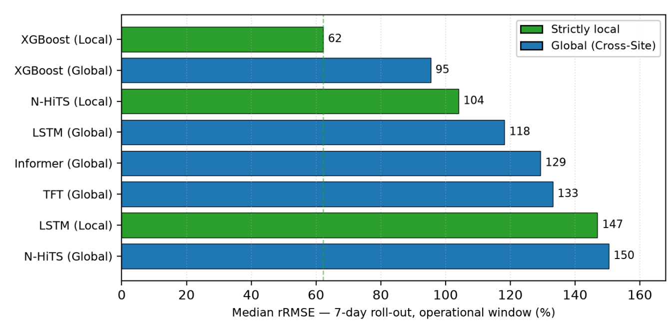

Fig. 3. Median rRMSE (7-day roll-out, operational window) per configuration. Local models in green, global Cross-Site models in blue; the dashed line marks the best local baseline.

and pinball loss; we report *diurnal* coverage (hours with $y > 0$) as the honest measure, since the physical night-zero forcing yields trivial nocturnal hits. Reported cells are medians over plants.

IV. Results

Results correspond to the *total*-scale run: LOPO over the 94 target plants, global models trained on the complete $N-1$ history. Table I reports the operational window; Fig. 3 visualizes the weekly roll-out ranking.

A. Point Accuracy

Three readings emerge from the weekly roll-out. First, **local XGBoost** is the most accurate configuration (62.2 % median rRMSE), followed by **global XGBoost** (95.4 %), the best Cross-Site model. Second, among the global deep networks the **LSTM** (118.2 %) surpasses the attention-based Informer (129.3 %) and TFT (133.1 %) and N-HiTS (150.4 %)—an ordering consistent with the caution that attention can be over-provisioned relative to simpler baselines [13]. Third, the degradation from *day1* to the weekly roll-out is moderate across all families, indicating that autoregressive median feedback does not destabilize the forecasts.

B. Commissioning-Ramp Window Sensitivity

Only 33 of the 94 plants exhibit a commissioning ramp longer than 24 h; for the rest the Raw and Operational windows coincide. On the aggregate median the two definitions differ only moderately, but on plants with a ramp the Raw window is

TABLE I
MEDIAN PERFORMANCE BY CONFIGURATION — OPERATIONAL WINDOW (n=33 PLANTS WITH A DISTINGUISHABLE COMMISSIONING RAMP). RRMSE IN %; DIURNAL COVERAGE OF THE 90 % INTERVAL; PINBALL P95 IN MWH. BEST VALUE PER COLUMN IN **BOLD**.

Configuration	Scope	rRMSE day1 (%)	rRMSE roll7d (%)	Cov_{90} diurnal	Pinball P95 (MWh)
XGBoost	Local	**40.6**	**62.2**	**0.86**	**0.033**
XGBoost	Global	77.5	95.4	0.99	0.091
N-HiTS	Local	73.6	104.0	0.66	0.107
LSTM	Global	103.4	118.2	0.54	0.140
Informer	Global	96.9	129.3	0.65	0.114
TFT	Global	110.2	133.1	0.44	0.123
LSTM	Local	102.6	146.9	0.79	0.211
N-HiTS	Global	130.6	150.4	0.36	0.277

TABLE II
MEDIAN RRMSE (%) OF THE 7-DAY ROLL-OUT BY SEASON IN WHICH THE EVALUATION WINDOW STARTS.

Model	Summer	Autumn	Winter	Spring
XGBoost local	40.5	35.9	114.8	101.5
XGBoost global	51.8	53.7	133.5	106.1
N-HiTS local	66.0	74.0	143.5	118.1
LSTM global	93.9	97.0	186.3	136.2
TFT global	87.7	102.0	168.8	130.9
Informer global	87.9	92.9	172.0	137.0
N-HiTS global	114.0	130.3	188.7	148.9

distorted: starting the evaluation over weeks of null generation drives the window mean toward zero and makes rRMSE undefined. Reporting both definitions is therefore deliberate—Raw is faithful to the historical record, whereas Operational yields interpretable metrics for a plant already in commercial operation and is the window used for the headline comparison.

C. Seasonal Sensitivity

For every plant we build Cold-Start windows starting in each of the four seasons, taken from its own post-commissioning history (independent of the actual connection season). Winter is **systematically the hardest season for every architecture** (Table II): in the weekly roll-out, global XGBoost rises from 51.8 % (summer) to 133.5 % (winter) and local XGBoost from 40.5 % to 114.8 %. Greater cloud variability and lower mean winter irradiance weaken the deterministic solar signal and amplify the relative weight of errors, with a direct operational implication: the forecast uncertainty of a newly connected plant depends strongly on the season of its commissioning.

D. Probabilistic Calibration

No paradigm calibrates its intervals ideally under the Cold-Start protocol, with deviations in opposite directions from the 90 % nominal (Table I, Cov_{90}). Evaluated on productive hours, the global deep networks are clearly **under-calibrated** (LSTM 0.54, Informer 0.65, TFT 0.44, N-HiTS 0.36): their bands are too narrow. Global XGBoost, at the opposite extreme, **over-covers** (0.99)—bands so wide they rarely fail but lose operational value, a defect symmetric to the networks', not a virtue. The closest to nominal is **local XGBoost** (0.86), whose superiority is confirmed by a much smaller pinball P95 (0.033 vs. 0.091 MWh for global). The networks' behaviour follows from the design: each series' local scaler is fit on the synthetic context, whose variance (an idealized solar profile) is below the real plant variability, producing narrow bands; the local deep baselines, which see real context, improve (LSTM local 0.79, N-HiTS local 0.66). Post-hoc recalibration (e.g., conformal prediction) is a clear line of future work.

E. Hypothesis Test

The hypothesis that the global multi-site model is *more accurate* than a locally trained model is **rejected in absolute terms**: no global model reaches the best local baseline. The single nuance is the matched-architecture comparison. (i) At equal deep architecture, global transfer is competitive but mixed: global LSTM beats local LSTM (118.2 % vs. 146.9 %), showing generalized climatic knowledge can exceed limited local history, yet local N-HiTS beats global N-HiTS (104.0 % vs. 150.4 %). (ii) Against the mature local baseline, local XGBoost (62.2 % roll-out; 40.6 % day1) dominates the best global model (95.4 %; 77.5 %). (iii) The operational value of the global model lies in *immediate availability*: the local paradigm needs months of accumulated data, whereas the global model forecasts usably from the first connected hour—the exact regime that motivates this work, and one in which no local competitor can exist.

V. DISCUSSION

The negative result is itself a contribution: it subjects the widespread belief in global attentional superiority to a stricter test than usual—target-blind, no local fine-tuning, against mature local baselines—and demarcates precisely the regime where Cross-Site transfer retains indisputable value: pure cold start. Several factors plausibly bound the global models' performance and define the roadmap. *No fine-tuning:* the global model ran strictly zero-shot while the local baselines that beat it observe real site data; a two-stage pretrain-then-finetune scheme should close much of the gap as history accrues. *Intermittency-aware architecture:* the models treat PV as a continuous signal, whereas it is intrinsically intermittent (deterministic night zeros, stochastic cloud gaps); explicit intermittency masks in the attention mechanism could redirect representational capacity to the productive signal. *Corpus scale and compute:* 94 plants and a 6 GB training budget may under-serve the higher-capacity TFT/Informer, offering an alternative explanation for their being outperformed by simpler models. *Interval recalibration* under transfer (conformal methods) is needed before operational deployment.

Transferable methodological findings. Beyond the benchmark, four lessons generalize to any cold-start forecasting study. (1) Commissioning ramps of null generation distort relative metrics unless the window is defined from sustained production, which motivated our Raw/Operational split. (2) Winter is systematically the hardest regime, doubling or tripling summer errors across all architectures. (3) Probabilistic calibration does not transfer to the deep networks under a synthetic context, whereas quantile gradient boosting calibrates well—a critical distinction for electricity-market use. (4) Target integrity—never imputing missing generation—is essential to metric honesty, in contrast to the common practice of zero-filling.

Limitations. The study is zero-shot by design (no fine-tuning arm); the rRMSE normalization inflates absolute magnitudes and is meaningful only in relative comparison; and operational MLOps deployment is out of scope.

VI. Conclusion

We benchmarked global Cross-Site transfer against strictly local baselines for cold-start PV forecasting under a target-blind LOPO protocol over 94 Chilean plants. In absolute accuracy the global-superiority hypothesis is rejected—a mature local XGBoost remains the model to beat—but at matched deep architecture global transfer is competitive, and in the pure cold-start regime it is the only paradigm that produces coherent, immediately available forecasts. Winter is the hardest season, and the deep networks' probabilistic intervals require recalibration under transfer. The contribution is not to crown an absolute winner but to quantify rigorously the trade-off between *immediate availability* and *asymptotic accuracy* that governs paradigm choice during the commissioning of a new solar plant.

Reproducibility

The medallion ETL, the LOPO split and all eight configurations are driven by a single orchestrator behind an automated test gate that halts the run on any failure; every metric is computed only over real observations on a shared canonical hourly grid, so the eight models are compared in a strictly paired fashion. Idempotent completion markers make long runs resumable, and the regional prior is recomputed inside each fold to preserve zero-shot integrity.

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